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Lithium recovery from shale gas produced water using solvent extraction☆

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ABSTRACT

Shale gas produced water is hypersaline wastewater generated after hydraulic fracturing. Since the produced water is a mixture of shale formation water and fracturing fluid, it contains various organic and inorganic components, including lithium, a useful resource for such industries as automobile and electronics. The produced water in the Marcellus shale area contains about 95 mg/L lithium on average. This study suggests a two-stage solvent extraction technique for lithium recovery from shale gas produced water, and determines the extraction mechanism of ions in each stage. All experiments were conducted using synthetic shale gas produced water. In the first-stage, which was designed for the removal of divalent cations, more than 94.4% of Ca^{2+}, Mg^{2+}, Sr^{2+}, and Ba^{2+} ions were removed by using 1.0 M di-(2-ethylhexyl) phosphoric acid (D2EHPA) as an extractant. In the second-stage, for lithium recovery, we could obtain a lithium extraction efficiency of 41.2% by using 1.5 M D2EHPA and 0.3 M tributyl phosphate (TBP). Lithium loss in the first-stage was 25.1%, and therefore, the total amount of lithium recovered at the end of the two-step extraction procedure was 30.8%. Through this study, lithium, one of the useful mineral resources, could be selectively recovered from the shale gas produced water and it would also reduce the wastewater treatment cost during the development of shale gas.

1. Introduction

Shale gas development has increased because of the development of hydraulic fracturing and horizontal drilling. Massive amounts of produced water are generated because each well needs 4–19 million liters of water injection during the hydraulic fracturing. After the process, 20–50% of the injected water flows back into the ground as produced water (International Energy Agency, 2012). Shale gas produced water is a mixture of the injected fracturing fluid and shale formation water; hence, there are various inorganic and organic components originated from the formation water and the injected fracturing fluid. Among the inorganic components, lithium is one of the useful resources that can be recovered. Lithium is a rare metal, and interest in its sources has been increasing because of its various uses such as lithium-based air treatment, ceramic and glass, grease, metallurgical, pharmaceutical, and polymer products (Jaskula, 2013). Lithium is a raw material used in aluminum-lithium alloys for both aerospace applications and rechargeable batteries. It has potential as a future nuclear fusion fuel (Nishihama et al., 2011). Lithium demand has increased, and will dramatically grow in the near future, especially because of the development of lithium ion batteries (Fig. 1). Finding new lithium sources is important to satisfy the increasing demands of Li.

Because of the large volume of seawater, several researchers are recovering lithium from seawater (Nishihama et al., 2011; Ryu et al., 2013). Lithium concentration in seawater is only 0.17 mg/L (Chitraker et al., 2001), whereas lithium content in the Marcellus shale area produced water is much greater (about 95 mg/L). Accordingly, lithium extraction from shale gas produced water is more efficient than that from seawater. Thus, shale gas produced water is considered as a new source of lithium. Several methods for lithium recovery from aqueous solutions have been studied, including electrodialysis (Hoshino, 2013), adsorption (Chitraker et al., 2001), electrochemical processes using the rechargeable battery principle (Kim et al., 2015), and solvent extraction (Kislik, 2012). Although the solvent extraction requires the use of a massive amount of organic solvent that can cause environmental pollution when they are not properly controlled (Kobayashi et al., 2010), the technique was chosen in this study because it is a

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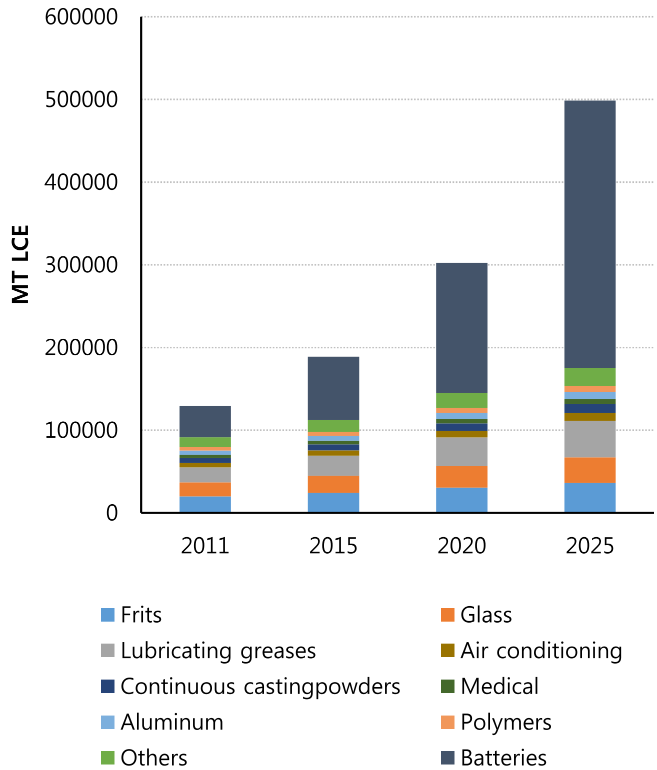

Fig. 1. Lithium consumption by application (Estimated) (Cabeza et al., 2015).

simple process with a short operation time (Yang et al., 2003). In addition, the solvent used in the process can be reused after stripping (Goodenough and Gaska, 1967), thus, making it an economical method. However, in solvent extraction, other cations in the produced water and the organic compounds in fracturing fluid such as glutaraldehyde (biocide), ethylene glycol (cross-linker), guar gum (gelling agent) and *N*, *N*-dimethyl formamide (corrosion inhibitor) can affect lithium extraction because of their affinity to solvent (Orem et al., 2014).

Many researchers have applied solvent extraction for alkali-metal recovery from various solutions (Umetani et al., 1987; Sadakane et al., 1975; Gabra, 1978; Hano et al., 1992). According to these studies, the active solvents for alkali-metal extraction are organic compounds that have a coordinating oxygen atom such as crown ethers, alcohols, and esters (Umetani et al., 1987). Crown ethers extract cations through an ion-pair extraction mechanism. One of the widely used macrocyclic polyether extractant is dibenzo-18-crown-6. It contains captured cations in the center of the macrocyclic ether ring, and forms crystalline complexes. This crown ether has the following cation affinity: $K^+ > Rb^+ > Cs^+ > Na^+ > Li^+$ (Sadakane et al., 1975). Selective separation of Li^+ from Na^+ using crown ethers is difficult since shale gas produced water contains much higher amounts of Na^+ than Li^+. Metal ion extraction using alcohols occurs via an electrochemical reaction. alcohols are highly polar and contain electronegative atoms; therefore, alcohols can react with electropositive cations. Gabra (1978) studied lithium extraction using n-butanol from a solution containing Na^+, K^+, and Ca^{2+}. The n-butanol extraction showed a higher selectivity for Li^+ than for Ca^{2+} (selectivity factor, $S^{Li}_{Ca} = 15$), but relatively low efficiency for Na^+ and K^+ ($S^{Li}_{Na} = 0.26$ and $S^{Li}_{K} = 0.32$). Similar to crown ethers, alcohols are not appropriate for separating Li^+ from Na^+. The ester-type extractant, di-(2-ethylhexyl) phosphoric acid (D2EHPA), forms a hydrophobic anion on deprotonation and chelates with cations in the aqueous phase. D2EHPA has a greater affinity for divalent cations than monovalent cations. Among K^+, Na^+, and Li^+, however, high Li^+ selectivity was observed (Hano et al., 1992).

The objectives of this study are to examine the applicability of the solvent extraction method for lithium recovery from diluted shale gas produced water and to reduce the effect of competitive cations and organic compounds on lithium recovery by using two-stage solvent extraction technique. It is expected, through this study, the useful resource including lithium is recovered and produced as a mineral resource during the treatment of shale gas produced water and thus the cost for the wastewater treatment in the shale gas production process could be reduced.

2. Theoretical background

2.1. Lithium extraction mechanism using D2EHPA

All experiments were carried out in this study by using D2EHPA as the extractant. D2EHPA is the most widely used extractant because it is chemically stable, has good loading and stripping characteristics, and has a low solubility in the aqueous phase (Ekberg and Petranikova, 2015).

D2EHPA deprotonation and anion formation proceeds by the following reaction:

$$(C_8H_{17}O)_2\overset{\overset{\displaystyle O}{\|}}{P}-OH \leftrightarrow (C_8H_{17}O)2\ \overset{\overset{\displaystyle O}{\|}}{P}O^- + H^+ \tag{1}$$

The resulting anion is hydrophobic that creates a cation chelate in the aqueous phase distributed between the aqueous and organic phases. Since D2EHPA exists predominantly as a dimer in nonpolar organic solvents, the general extraction reaction formula can be written as follows:

$$M^{n+} + \frac{n+x}{2}\ \overline{(HR)_2} \leftrightarrow \overline{MR_n\ (HR)_x} + nH^+ \tag{2}$$

$$K_e = \frac{\left[MR_n\ (HR)_x\right]\ \left[H^+\right]^n}{\left[M^{n+}\right]\left[(HR)_2\right]^{\frac{n+x}{2}}} = \frac{D_M\ \left[H^+\right]^n}{\left[(HR)_2\right]^{\frac{n+x}{2}}} \tag{3}$$

where M is the metal cation, n is the metal cation charge, HR is D2EHPA, and x is the solvation number of the complex. In equation (3), K_e is the extraction equilibrium constant from equation (2), and D_M is the distribution ratio, which can be calculated by equation (4).

$$D_M = \frac{\left[MR_n\ (HR)_x\right]}{\left[M^{n+}\right]} = \frac{\text{Concentration of metal ion species in the organic phase}}{\text{Concentration of metal ion species in the aqueous phase}} \tag{4}$$

After rearranging equation (3), the following formula can be obtained:

$$\log D_M = \frac{n+x}{2}\ \log\left[(HR)_2\right] + \ \log\frac{K_e}{\left[H^+\right]^n} \tag{5}$$

The separation factor, S, of each cation M_1 form another cation M_2 in the extraction was calculated by the equation below (Gabra, 1978).

$$S_{M_2}^{M_1} = \frac{Ratio\ of\ M_1\ ion\ in\ the\ organic\ phase/M_1\ ion\ in\ the\ aqueous\ phase}{Ratio\ of\ metal\ ion\ M_2\ in\ the\ ortanic\ phase/M_2\ ion\ in\ the\ aqueous\ phase} = \frac{D_{M1}}{D_{M2}} \tag{6}$$

Fig. 2 illustrates the logarithmic distribution ratio of Li^+ in different D2EHPA concentrations. As shown in equation (5), the slope is denoted as $(n + x)/2$. Considering that $n = 1$ for Li^+, the solvation number, x, was calculated as approximately two $(x = 2.12)$.

Consequentially, the lithium-D2EHPA extraction reaction formula is obtained as follows:

$$Li^+ + 1.56[(RO)_2\overset{\overset{O}{\|}}{P}-OH]_2 \leftrightarrow [(RO)_2\overset{\overset{O}{\|}}{P}O]Li[HO\overset{\overset{O}{\|}}{P}(RO)_2]_{2.12} + H^+ \tag{7}$$

2.2. Synergistic effect of tributyl phosphate (TBP)

When organophosphorus acids such as D2EHPA are used as the extractants, TBP generally acts as a synergistic additive (Shamsuddin, 2016). Addition of the proper amount of TBP significantly improves the extractability of metal ions (Hano et al., 1992). The synergistic effect (especially for Li^+) can be explained by the following reaction.

$$LiR\ xHR + TBP \leftrightarrow LiR\ (x-1)HRTBP + HR \tag{8}$$

TBP displaces one molecule of D2EHPA in the Li-D2EHPA complex, creating one molecule of D2EHPA that can react with another Li^+. As expressed in equation (8), the lithium distribution ratio, D_{Li}, increases with increasing TBP concentrations. Addition of excess TBP, however, suppresses lithium extraction, as per equation (9).

$$HR + TBP \leftrightarrow HRTBP \tag{9}$$

Excess TBP reacts with D2EHPA, thereby decreasing the available amount of D2EHPA for reaction with Li^+. Therefore, the use of an appropriate amount of TBP is critical for lithium extraction.

3. Materials and methods

3.1. Sample preparation

Shale gas produced water was synthesized to have the same constituent concentrations as the Marcellus shale produced water reported previously (Haluszczak et al., 2012). Since shale gas produced water consists of both the fracturing fluid and shale formation water, it was assumed that the same volume of the fracturing fluid and formation water was contained in the produced water. The produced water was prepared mainly with cations that are competitive with lithium in liquid-extraction, including Ca^{2+}, Li^+, Mg^{2+}, Na^+, Sr^{2+}, and Ba^{2+}. First, the synthetic fracturing fluid was

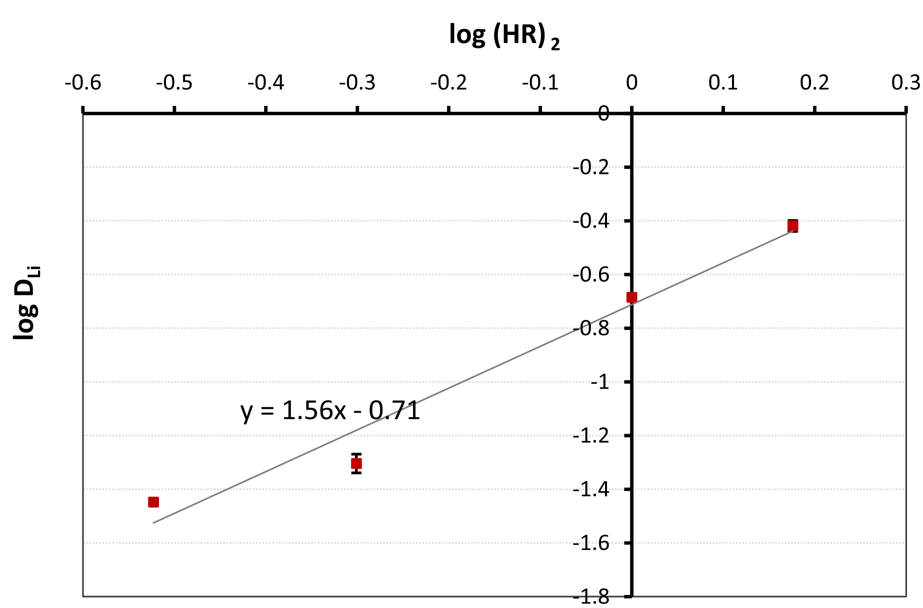

Fig. 2. Graphical analysis for calculation of solvation number in Li-D2EHPA reaction.

Table 1
Fracturing fluid additives and main compounds (Spellman, 2012).

Additive type	Volumetric composition[a], %	Main component	Purpose
Diluted acid (15%)	0.12	Hydrochloric acid	Helps dissolve minerals and initiates cracks in the rock
Biocide	0.001	Glutaraldehyde	Eliminates bacteria in the water
Breaker	0.01	Ammonium persulfate	Allows delayed breakdown of the gel polymer chains
Corrosion inhibitor	0.002	*N*, *N*-dimethyl formamide	Prevents corrosion of the pipe
Cross-linker	0.007	Borate salts	Maintain friction between the fluid and the pipe
Friction reducer	0.09	Polyacrylamide	Minimizes friction between the fluid and the pipe
Gel	0.056	Guar gum	Thickens the water in order to suspend the sand
Iron control	0.004	Citric acid	Prevents precipitation of metal oxides
KCl	0.06	Potassium chloride	Creates a brine carrier fluid
pH adjusting agent	0.011	Sodium carbonate	Maintains effectiveness of other components
Proppant	8.95	Silica, quartz sand	Allows fractures to remain open so gas can escape
Scale inhibitor	0.043	Ethylene glycol	Prevents scale deposits in pipes
Surfactant	0.085	Isopropanol	Used to increase viscosity of the fracturing fluid

[a] Source: US Department of Energy, 2009.

charged with the main compounds given in Table 1. Proppants are solid materials that are used to keep fractures open after pressure reduction; 40/70 mesh grain sand is commonly used in the Marcellus area (Beckwith, 2011). Therefore, 40/70 mesh (212–420 μm) silica particulates were added into the synthetic fracturing fluid. Second, various chemicals such as LiBr solution and metal chlorides, were added to the synthetic fracturing fluid to meet the concentrations of these compounds in real shale gas produced water. Third, the pH was adjusted to the real produced water pH range (5.8–6.6) using 1 N NaOH. Finally, the produced water was allowed to settle for 24 h for the precipitation of particulates and guar gum. The resultant supernatant liquid was used as the shale gas produced water sample.

3.2. Experiment

In all extraction experiments, D2EHPA (97%, Sigma-Aldrich, USA) was used as an extractant, and kerosene (Sigma-Aldrich, USA) was used as a diluent solvent to match the concentration of extractant. The organic/aqueous phase ratio was 1:1. The two-phase solutions were agitated in a separating funnel, using a shaking incubator (SH-BSI16R, Samheung Instrument, Korea), at a constant speed and temperature (150 rpm and 25 °C, respectively) for 30 min. The resultant mixture was allowed to equilibrate until the two phases reached equilibrium, after which the aqueous solution was withdrawn. Basic properties of the aqueous phase such as pH and conductivity were measured using a multimeter (Orion Star A329, Thermo Fisher Scientific, USA). The aqueous solution was filtered through a 0.45 μm polytetrafluoroethylene (PTFE) filter (Millipore, Germany) to analyze constituent concentrations by inductively coupled plasma optical emission spectrometry (ICP-OES, Optima 8300, PerkinElmer, USA) and ion chromatography (IC, Dionex ICS-1100, Thermo Scientific, USA).

3.2.1. First-stage solvent extraction

The first-stage solvent extraction experiments were performed to reduce the competitive effects of divalent cations, especially Ca^{2+}, Mg^{2+}, Sr^{2+}, and Ba^{2+} that have a greater affinity than Li^{+} for the extractant. The purpose of this extraction stage was to remove >90% of the divalent cations in the produced water. Four different extractant concentrations (0.3, 0.5, 1.0, and 1.5 M) were prepared by diluting D2EHPA in kerosene. Since the total dissolved solids (TDS) level of shale gas produced water is very high (up to 200,000 mg/L), the concentrations of other cations that suppress lithium recovery were high. Extraction efficiency of all cations decreases with high TDS levels, so the first-stage extraction experiments were conducted with various shale gas produced water dilution rates (1, 10, 25, and 50×) to determine the most efficient TDS level for cation extraction.

3.2.2. Second-stage solvent extraction

The purpose of the second-stage extraction was for selective recovery of Li^{+}. The organic phase was prepared by diluting D2EHPA and synergistic additive TBP in kerosene. The aqueous phases from the first-stage extraction were applied to the second-stage extraction experiments. D2EHPA (1.5 M) was used as an extractant with different TBP concentrations (0.1, 0.3, 0.5, 1.0, and 1.5 M).

4. Results and discussion

4.1. Composition of synthetic shale gas produced water

The synthetic produced water was analyzed, and the concentrations of the components are shown in Table 2. The second column shows the concentration ranges of components in the synthetic produced water. The third column shows the values of real shale gas produced water from the Marcellus area which were reported previously (Haluszczak et al., 2012).

Most of the values in the synthetic produced water are within the range of real flowback water, except alkalinity. The pH and alkalinity were adjusted by addition of sodium hydroxide and sodium carbonate, but the alkalinity was still lower than expected. This is due to the addition of hydrochloric acid in the synthetic fracturing fluid.

4.2. Calculation of extraction efficiency

The cation concentrations in the organic phase were calculated by a mass balance equation using their initial and final aqueous phase concentrations. Then the extraction efficiency was determined by the following equation:

$$\text{Extraction efficiency \%} = \frac{\text{Equilibrium concentation of metal ion in the organic phase}}{\text{Initial concentration of metal ion in the aqueous phase}} \times 100 \qquad (10)$$

Table 2
Composition of synthetic and real shale gas produced water.

	Synthetic produced water [mg/L]	Flowback water [mg/L]
Total dissolved solids	9400–103,180	3010–228,000
pH	5.86–6.8	5.8–6.6
Alkalinity [mg $CaCO_3$/L]	7–8	26–95
Ca	10,310–12,710	204–14,800
Li	97.7–106	4–202
Mg	856–947	22–1800
Na	24,320–29,780	1100–44,100
Sr	2301–2601	46–5350
Ba	1780–1924	76–13,600
Cl	66,778–73,891	1070–151,000
Br	998–1172	16–1190

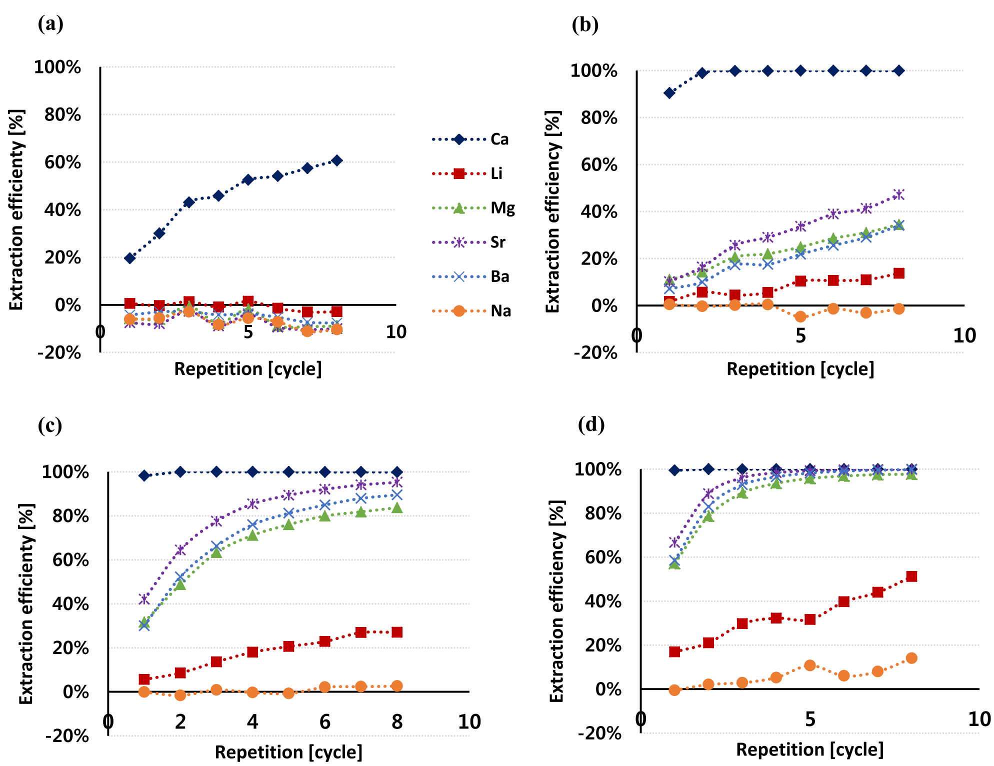

Fig. 3. Cations extraction efficiency in different dilution rates of shale gas produced water: (a) 1× (b) 10×, (c) 25×, (d) 50×.

4.3. First-stage extraction for divalent cation removal

4.3.1. Cation extraction characteristics with different shale gas produced water dilution rates

Repetitive extractions were performed up to eight times to improve the divalent cation removal efficiency in the produced water. Preliminary tests were conducted to determine the most efficient extractant concentration. From these results, 1.5 M D2EHPA was chosen as the concentration for the first-stage extraction. Fig. 3(a) shows the results of eight extraction cycles on the synthetic shale gas produced water. Only Ca^{2+} was extracted, and its extractability increased with repetition until reaching 60.6%. The initial Ca^{2+} concentration was 10,310 mg/L, and the final aqueous solution concentration after the eightfold repetition was 4059 mg/L. The other five ions were not extracted possibly due to the high TDS level of the shale gas produced water.

A previous lithium extraction study reported that the TDS of geothermal water for the lithium recovery is about 2255 mg/L (Hano et al., 1992). Synthetic shale gas produced water has TDS = 100,000 mg/L on average; around 50 times higher than geothermal water. Therefore, the repetitive extraction tests were conducted with a gradual increase in produced water dilution rates (10, 25, and 50×) to find the most favorable dilution rate. In the case of 10× dilution (Fig. 3(b)), after eight extraction cycles, almost all of the Ca^{2+} was removed (99.9%). However, the five other cations still showed relatively low extractability. The extraction efficiency of Mg^{2+}, Sr^{2+}, and Ba^{2+} were 34.5%, 47.2%, and 34.1%, respectively. In the 25× dilution (Fig. 3(c)), all of the Ca^{2+} was removed, and the extractabilities of Mg^{2+}, Sr^{2+}, and Ba^{2+} were 83.7%, 95.2%, and 89.5%, respectively. Although the extraction efficiency of divalent cations increased in the 25× dilution test, in the cases of 10× and 25× dilution, the final concentrations of Mg^{2+}, Sr^{2+}, and Ba^{2+} were still higher than Li^{+} (Table 3). Thus, these dilution rates are thought to be still low for the complete removal of divalent cations from the shale gas produced water. Fig. 3(d) shows the extraction results of 50× dilution. After a single extraction, 99.4% of the Ca^{2+} and more than 50% of the other cations were extracted. After eight cycles, all of the Ca^{2+}, Sr^{2+}, and Ba^{2+}, and 97.9% of the Mg^{2+} ions were

Table 3
Concentrations of components in the aqueous phase after eight repetitive extractions.

	Initial Concentration [mg/L]	Final concentration in aqueous phase [mg/L]			
		1×	10×	25×	50×
Ca^{2+}	10,310	4059	0.77	0.69	<D.L.
Sr^{2+}	2301	2532	124	4.19	<D.L.
Ba^{2+}	1789	1926	117	7.03	<D.L.
Mg^{2+}	856	932	56.3	5.25	0.41
Li^{+}	97.7	101	8.22	2.65	0.97
Na^{+}	24,320	26,845	2533	946	437

*D.L. means detection limit.

removed. In the case of Li^+, the extractability increased until 51.1%. However, when compared to the final aqueous phase concentration of each component, the concentrations of all of divalent cations were lower than that of lithium.

Determining the proper dilution rate was very important for the recovery efficiency of lithium in the two-stage extraction. Although the removal rates of competitive cations such as Mg^{2+}, Sr^{2+}, and Ba^{2+} were significantly higher than that of Li^+ in the 25× dilution test result, concentrations of those cations in the aqueous solution were higher than the lithium ion concentration. Because the divalent cations have greater affinity to the extractant, D2EHPA, than lithium ion, the competitive extraction of the Mg^{2+}, Sr^{2+}, and Ba^{2+} would decrease the extraction efficiency of Li^+ in the second stage. Therefore, the dilution rate in which the concentrations of the divalent cations in aqueous solution after the first-stage were less than the lithium concentration was selected. Consequentially, the most appropriate dilution rate was 50×.

4.3.2. *Repetitive extractions with different concentrations of D2EHPA*

The 50× dilution rate from the previous section was chosen for these solvent extraction tests. The extractability of divalent cations was high with 1.5 M D2EHPA, but lithium loss was relatively high as well. Therefore, different concentrations of D2EHPA (0.3, 0.5, 1.0, and 1.5 M) were applied to select the most efficient extractant concentration, as well as the appropriate number of repetitions. Almost all of the Ca^{2+} ions were removed at all D2EHPA concentrations, but the extractability of other divalent cations decreased at 0.3 and 0.5 M of D2EHPA, despite the fact that the extraction was conducted two times more than at 1.0 and 1.5 M of D2EHPA (Fig. 4). When compared to the results shown in Fig. 4(c–d), extraction at 1.0 M D2EHPA is more effective. When the removal efficiencies for Mg^{2+}, Sr^{2+}, and Ba^{2+} were about 94.4%, 98%, and 97% after eight cycles for 1.0 M of D2EHPA and four cycles for 1.5 M of D2EHPA, lithium loss was 23.2% and 32.2% for 1.0 M and 1.5 M D2EHPA, respectively. Therefore, eight extraction cycles using 1.0 M D2EHPA was chosen to remove the divalent cations in the first-stage extraction. The loss of lithium did not increase after six cycles, but for the other divalent cations, removal efficiency increased with increasing extraction cycles. The extractabilities of Ca^{2+}, Mg^{2+}, Sr^{2+}, and Ba^{2+} were 99.9%, 93.7%, 99.2%, and 97.2%, respectively after eight cycles. Also, the separation factors of the divalent cations from lithium ion were significantly high ($S_{Li}^{Ca} = 10398$, $S_{Li}^{Sr} = 405$, $S_{Li}^{Ba} = 124$, $S_{Li}^{Mg} = 50$). The aqueous solution obtained after the eight-cycle extraction using 1.0 M D2EHPA was denoted as "Solution A".

4.4. *Second-stage extraction for selective lithium recovery*

Extraction experiments were performed to recover lithium from Solution A by addition of synergistic additive TBP to extractant D2EHPA. An appropriate amount of TBP is important to improve the lithium extraction efficiency. Lithium extraction is inhibited with excess TBP addition (equation (9)). Therefore, to determine the proper TBP concentration, repetitive extraction tests were designed with various TBP concentrations (0.1, 0.3, 0.5, 1.0, and 1.5 M) using 1.5 M D2EHPA (Fig. 5). The extractability of all ions decreased with increased TBP concentration. The extraction efficiencies for Sr^{2+} and Ba^{2+} were higher than for lithium and >98% of both ions were removed in the first-stage extraction. Thus, their concentrations in the second-stage extraction were very low compared to the concentration of lithium. For Mg^{2+} and Na^+, the extractability decreased negatively with an increase in TBP concentration. This

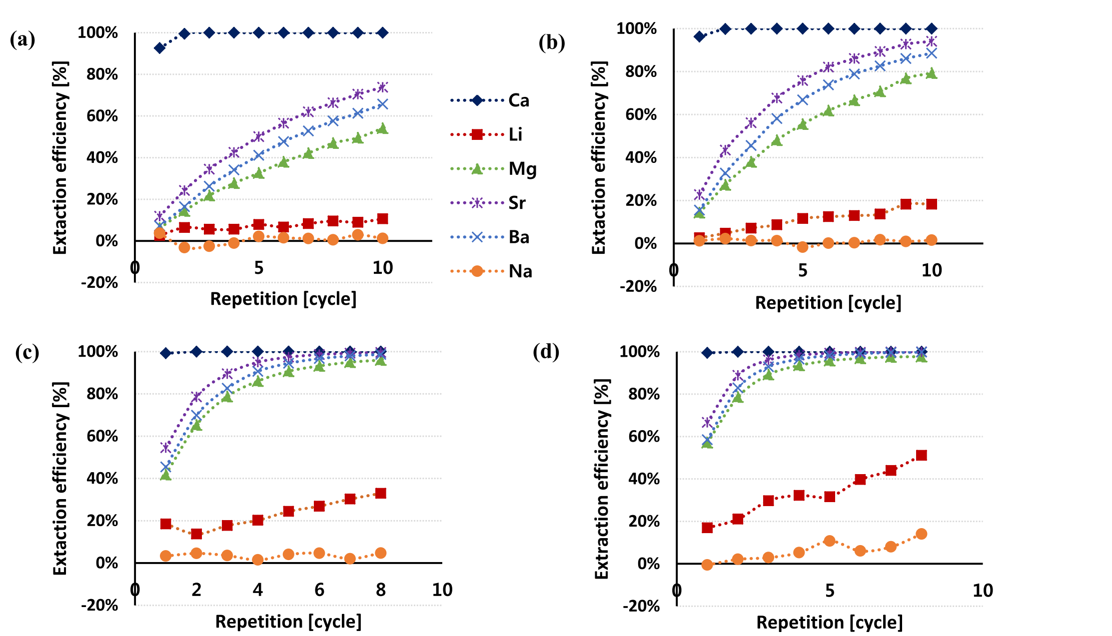

Fig. 4. Repetitive extractability from 50× diluted shale gas produced water with different D2EHPA concentrations: (a) 0.3 M, (b) 0.5 M, (c) 1.0 M, (d) 1.5 M.

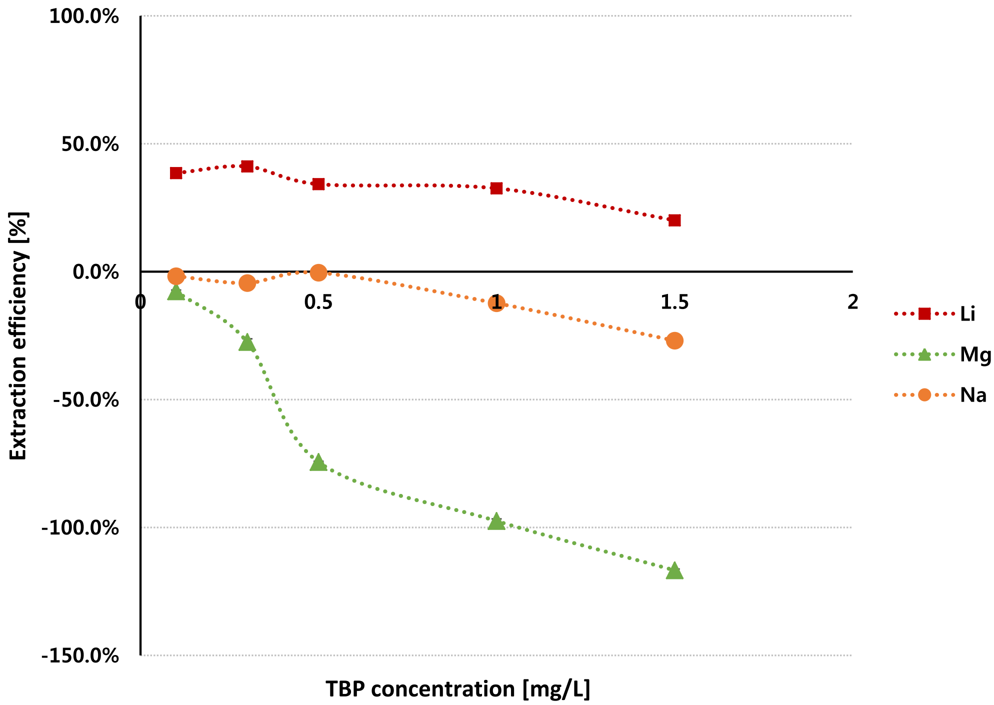

Fig. 5. Cations extraction efficiency in solution A with different TBP concentrations (The data for 0.1M, 0.3M, 0.5M and 1.0M were extraction results after 10 times repetition, whereas 1.5M was extraction result after 8 times repetition because of experimental problem).

means that the aqueous phase concentrations of these ions increased, so it was assumed that TBP addition affects the dissolution of Mg^{2+} and Na^+ in Solution A. For Li^+ extraction, the extraction efficiencies after ten repetitions with 0.1, 0.3, 0.5, and 1.0 M TBP, and eight repetitions with 1.5 M TBP were 38.6%, 41.2%, 34.2%, 32.6%, and 20.1%, respectively. The highest efficiency was observed with 0.3 M TBP. The calculated separation factor of lithium from sodium was -16.41 in the second stage due to the negative extraction efficiency sodium. Considering first and second stage extraction, however, the separation factor of lithium from sodium was 30.7. After ten cycles with 0.3 M TBP, 41.2% of the Li^+ was recovered. In the first-stage extraction, lithium loss was 25.1%, therefore, after the two-step extraction the total amount of Li^+ recovered was 30.8%.

5. Conclusions

Lithium recovery experiments were performed to determine the applicability of solvent extraction techniques on shale gas produced water. Because of the high TDS of the produced water, the cation extractability was low. Therefore, 1, 10, 20, and 50× dilution rates were applied to the solvent extraction tests, with 50× determined as the optimal dilution rate. In all of the experiments, the extractant D2EHPA showed the following affinity: $Ca^{2+} > Sr^{2+} > Ba^{2+} > Mg^{2+} > Li^+ > Na^+$. This means that lithium extraction efficiency is lower than that for the divalent cations. In order to both reduce the competitive impact of the divalent ions on lithium recovery and improve the selectivity of lithium extraction, a two-stage solvent extraction was suggested. In the first-stage, repetitive extraction was tested, and we found that eight repetition cycles using 1.0 M D2EHPA was the most efficient process. The aqueous solution obtained after eight extraction cycles with 1.0 M D2EHPA was designated as "Solution A". In the second-stage, lithium extraction was conducted with Solution A, with the addition of TBP to improve lithium extractability. Five different TBP concentrations (0.1, 0.3, 0.5, 1.0, and 1.5 M) with 1.5 M D2EHPA were applied to determine the optimal concentration. The highest lithium extraction efficiency (41.2%) was observed using 1.5 M D2EHPA and 0.3 M TBP. During the first-stage extraction, more than 94% of all divalent cations were removed with high distribution ratio compared to the lithium. D2EHPA has a higher affinity for Li^+ than other monovalent cations. Therefore, Li^+ was recovered with high separation factor from sodium. There are limitations in scaling-up the solvent extraction process using a large amount of organic solvent in a batch-type reactor. Further studies about continuous flow solvent extraction process and stripping for the reuse of solvent and extractant are necessary to apply two-stage solvent extraction technique for the field application.

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